

Linear Regression on Synthetic Data: A Minimal Reproducible Example

AutoReproducer Sample Paper

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Abstract

This paper presents a minimal reproducible example of ordinary least squares (OLS) linear regression on a synthetic dataset. The goal is to provide a clean baseline with a single algorithm, a single metric, and minimal dependencies, suitable for automated reproduction pipelines.

1. Method

We use ordinary least squares (OLS) to fit a linear model $y = Xw + b$. Given a design matrix X in $\mathbb{R}^{(n \times d)}$ and target vector y in $\mathbb{R}^n$, the closed-form solution is obtained by appending a column of ones to X and solving the normal equations.

Formally, let $X_b = [X, 1]$ in $\mathbb{R}^{(n \times (d+1))}$. The optimal parameters are:

$$w_b = (X_b^T X_b)^{-1} X_b^T y$$

where $w_b[-1]$ is the bias term b .

2. Experimental Setup

Dataset: synthetic, generated with `numpy.random.randn`.

- Samples: $n = 1000$
- Features: $d = 2$
- Ground-truth weights: $w = [2.5, -1.8]$
- Ground-truth bias: $b = 3.0$
- Noise: Gaussian with $\text{std} = 0.3$
- Random seed: 42 (for reproducibility)

Evaluation metric:

- Mean Squared Error (MSE) on the same synthetic dataset.
- Expected value: MSE approximately 0.09 (depends on random noise).

Dependencies:

- Python 3.11+
- numpy

3. Implementation (Python)

```
import numpy as np
```

```
# reproducible synthetic data
np.random.seed(42)
n_samples = 1000
```

```

n_features = 2
X = np.random.randn(n_samples, n_features)
true_w = np.array([2.5, -1.8])
true_b = 3.0
noise = np.random.randn(n_samples) * 0.3
y = X @ true_w + true_b + noise

# OLS closed-form solution
X_b = np.c_[X, np.ones(n_samples)]
w_b = np.linalg.pinv(X_b.T @ X_b) @ X_b.T @ y
w_pred = w_b[:-1]
b_pred = w_b[-1]

# validation
y_pred = X @ w_pred + b_pred
mse = float(np.mean((y - y_pred) ** 2))

print(f"MSE={mse:.4f}")
print(f"w_pred={w_pred.round(4).tolist()}")
print(f"b_pred={b_pred:.4f}")

```

4. Results

Running the code above on a standard machine (Python 3.11, numpy 1.26+) produces the following expected output:

```

MSE=0.0892
w_pred=[2.5074, -1.7963]
b_pred=2.9876

```

Note: the exact MSE may vary slightly (+/-0.01) due to floating-point differences across numpy versions and platforms, but should remain below 0.15.

5. Conclusion

This minimal example demonstrates a fully reproducible linear regression pipeline. With only numpy as a dependency and a single evaluation metric (MSE), it serves as an ideal sanity-check for automated reproduction systems.